

Hongfu Sun, PhD ([Google Scholar](#))

[Senior Lecturer \(permanent\)](#) | University of Newcastle | Australia | hongfu.sun@newcastle.edu.au

[Honorary Senior Research Fellow](#) | University of Queensland | Australia | hongfu.sun@uq.edu.au

ACADEMIC BACKGROUND

- **Senior Lecturer (permanent)**, University of Newcastle, Newcastle, Australia 2024/02 – Present
- **DECRA Senior Research Fellow**, University of Queensland, Brisbane, Australia 2023/01 – 2024/02
- **DECRA Research Fellow**, University of Queensland, Brisbane, Australia 2019/03 – 2022/12
- **Postdoctoral Researcher**, University of Calgary, Calgary, Canada 2015/11 – 2019/03
- **Doctor of Philosophy**, University of Alberta, Edmonton, Canada 2010/09 – 2015/11
- **Bachelor of Science**, Huazhong University of Science and Technology, China 2005/09 – 2019/06

LEADERSHIP AND SUPERVISION

- Discipline Lead of Medical Engineering Research Centre (Uni. Newcastle and HMRI) 2024
- Graduated 3 and currently taking 4 PhD students as the Principal Supervisor (Uni. Queensland) 2019 – Present
- Graduated 6 Master by Coursework and 2 Honours students (Uni. Queensland) 2020 – 2023

RESEARCH THEMES

- Medical imaging, MRI, quantitative MRI, Novel imaging methods
- Artificial Intelligence (AI), Machine/Deep learning, Generative models

TEACHING

- 2026- (Uni. Newcastle/Singapore) – Course Coordinator, **ELEC4401: Signal Processing**
- 2024- (Uni. Newcastle/Singapore) – Course Coordinator, **ENGG1003: Procedural Programming** [600+ students]
- 2024- (Uni. Newcastle) – Course Coordinator, **MENG3451: Medical Imaging and Signal Processing**
- 2019-2023 (Uni. Queensland) – Course Coordinator, **BIOE6601: Medical Imaging** [average rating 4.8/5.0]

HONORS AND AWARDS

- **Dr F & Mrs ME Zaccari Scholarship** for Schizophrenia research (1 PhD scholarship across UQ) 2023
- **UQ Knowledge Exchange & Translation Award** (2 awardees from Engineering Faculty) 2022
- **Nominee as President** of ISMRM Australia and New Zealand community 2022
- **Collaboration Award** from UQAI (6 awardees from Engineering Faculty) 2022
- **Chair** of ISMRM Data Processing Scientific oral and poster sessions 2022
- **Chair** of ISMRM QSM Scientific oral and poster sessions 2021
- **Chair** of ISMRM-ANZ acquisition and reconstruction session 2021
- **ARC Discovery Early Career Researcher Award (DECRA)** (the most prestigious government early-career fellowship award with only 200 each year across Australia) 2021
- **The “Rising Star” Award** from School of ITEE (4 awardees from School) 2021
- **Early Career Researcher Award** from UQ 2020
- **Cover Article Award** for MRM (top 1 MRI journal) January 2020 issue 2020
- **Summa Cum Laude Merit Award** from ISMRM (top 3% of 6000+ abstracts) 2014/2015/2019
- **Conference Travel Award** from Hotchkiss Brain Institute 2018
- **Distinguished Reviewer Award** for MRM (top 1 MRI journal) 2017/2018
- **Distinguished Reviewer Award** for Journal of Magnetic Resonance Imaging 2016-2018
- **Postgraduate Fellowships in Health Innovation** (the most prestigious government fellowship with 20 awardees across Alberta province in Canada) from Alberta Innovates 2016-2018
- **Trainee Stipend Award** from ISMRM 2013-2015/2018
- **Research Exchange Award** from HBI-Melbourne Exchange Program (2 awardees from Uni.) 2017
- **Chair** of ISMRM QSM Educational oral and Scientific oral sessions 2017
- **Postdoctoral Fellowship** from T. Chen Fong in Medical Imaging Science (1 awardee from Uni.) 2016
- **Postdoctoral Fellowships** from Harley Hotchkiss - Samuel Weiss 2015
- **Cover Article Award** for MRM (top 1 MRI journal) May 2015 issue 2015
- **NIH Young Investigator Travel Award** for QSM International Workshop 2014/2015
- **Travel Award** from University of Alberta Graduate Student Association 2014
- **Consortium Ambassador** for Canadian Graduate Engineering 2013/2014
- **Scholarship** from Faculty of Medicine & Dentistry 75th Anniversary (\$7000) 2012
- **Scholarship** from the University of Alberta BME Graduate Program (\$6500) 2011

SUCCESSFUL GRANTS (CI stands for Chief Investigator, equivalent to Principal Investigator (PI) in North America system, A-C indicates the position in the CI list)

• University of Newcastle Startup Grants	\$32,500	Sole CI	2024-2025
• Dr F & Mrs ME Zaccari Scholarship	\$240,000	Sole CI	2023-2026
• <u>NHMRC Ideas Grants</u>	\$611,098	CI C	2024-2026
• <u>ARC Discovery Projects (DP)</u>	\$512,607	CI C	2023-2025
• UQ Knowledge Exchange & Translation Award	\$139,124	Sole CI	2023
• <u>ARC Discovery Early Career Researcher Award (DECRA)</u>	\$429,000	Sole CI	2021-2023
• UQAI Collaboratory ECR Seed Fund	\$20,000	CI A	2022
• ITEE Research Support Funding	\$10,000	Sole CI	2021
• EAIT Early Career Researcher Philanthropic Grants	\$33,000	Sole CI	2021
• ITEE Startup Grants	\$50,000	Sole CI	2019-2021
• UQ Early Career Researcher Grants	\$25,875	Sole CI	2020
• <u>Alberta Innovates Postgraduate Fellowships in Health Innovation</u>	\$165,000	Sole CI	2016-2018
• RHISE HBI Melbourne Trainee Research Exchange	\$14,000	Sole CI	2017
• T. Chen Fong Postdoctoral Fellowship in Medical Imaging Science	\$60,000	Sole CI	2016
• Harley N. Hotchkiss Postdoctoral Recruitment Fellowship	\$50,000	Sole CI	2016

COMMUNITY AND PROFESSIONAL ACTIVITIES

• <i>Host</i> National Youth Science Forum of University of Newcastle visit (20 students)	2024/07
• <i>Staff Representative</i> for Uni. Newcastle Open Day of the Bachelor of Medical Engineering	2024/08
• <i>Outreach</i> to Newcastle Grammar School Medical Engineering Program (16 Y11 students)	2024/09
• <i>Guest Editor</i> of NeuroImage, Frontiers in Radiology, Frontiers in Neuroscience, Information	2022-2024
• <i>HDR Scholarship Ranking Committee</i> for the School of EECS	2022-2024
• <i>Panel Reviewers Committee</i> of Low and Negligible Risk (LNR) ethics for UQ	2021-2024
• <i>HDR Milestone Panel Member</i> for 20+ PhD students from the School of EECS, CAI, and QBI	2020-2024
• <i>Grant Review</i> for Natural Sciences and Engineering Research Council of Canada Doctoral Award 2017; The Austrian Science Fund 2022; ARC DECRA 2022; Fonds de Recherche du Québec – Nature et technologies 2023; ARC Linkage 2023	2017-2023
• <i>Peer Review</i> for MICCAI, ISMRM; IEEE; Radiology; NeuroImage; MRM; JMRI; NMR in Biomedicine; Frontiers; etc. (130+ papers recorded by Web of Science)	2017-Present
• <i>Organizing Committee</i> of “MRI Together 2022” International Workshop (1000+ attendees)	2022
• <i>Staff Representative</i> at UQ Open Day and EAIT Experience Day for Biomedical Engineering	2020-2022
• <i>Chair and Moderator</i> for ISMRM conference (6000+ attendees)	2017-Present
• <i>Invited Editorial</i> for Journal of Magnetic Resonance Imaging	2019/12
• <i>Academic Exchange</i> at Melbourne Brain Centre Imaging Unit, Australia	2017/05-07

PEER-REVIEWED PUBLICATIONS (*10 as first author, 18 as last/corresponding author **)

1. Z. Xiong, Y. Gao, F. Liu, D. Emery, K. Butcher, A.H. Wilman, **H. Sun***; Advancing Quantitative Susceptibility Mapping With 2.5D Diffusion Models for Rapid Intracranial Hemorrhage Quantification; Magn. Reson. Med. 2026.
2. S. Prima, Z. Xiong, A.H. Wilman, **H. Sun***; DeepRelaxo: Fast Mono-Exponential Magnitude Brain R2* Mapping with Reduced Echoes Using Self-Supervised Deep Learning; Magn. Reson. Med. 2026.
3. T. Ding, H. Chen, Y. Gao, F. Liu, M. A. Cloos, **H. Sun***; Adaptive Gate-Aware Mamba Networks for Magnetic Resonance Fingerprinting; 2026 IEEE International Symposium on Biomedical Imaging (ISBI), London, UK, April 2026.
4. J. Liu, Q. Lin, Z. Xiong, S. Shan, C. Liu, M. Li, F. Liu, G. Bruce Pike, **H. Sun**, Y. Gao; Highly Undersampled MRI Reconstruction via a Single Posterior Sampling of Diffusion Models; IEEE Transactions on Medical Imaging. 2026; doi: 10.1109/TMI.2026.3654585.
5. K. T. Islam, S. Dayarathna, S. Zhong, P. Zakavi, H. Kavvounias, S. Farquharson, G. Durbridge, **H. Sun**, S. Bacchi, G. F. Egan, M. Barth, A. Dwyer, K. L. McMahon, P. M. Parizel, M. Law, Z. Chen; SynPoC: a high-quality generative diffusion model for transforming ultra-low-field point-of-care MRI using high-field MRI representations; Scientific Reports; 2026;16(1):3285; doi: 10.1038/s41598-025-33162-9.
6. R. Jin, C. Chen, Y. Liu, P. Wu, **H. Sun**, R. Zheng, M. Zeng, M. Li, Y. Gao; GateFuseNet: An Adaptive 3D Multimodal Neuroimaging Fusion Network for Parkinson’s Disease Diagnosis; 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM); doi: 10.1109/BIBM66473.2025.11356155.
7. H. Askari, Y. Luo, **H. Sun***, F. Roosta; Latent Refinement via Flow Matching for Training-free Linear Inverse Problem Solving; NeurIPS 2025.
8. **H. Sun**; Editorial for "Accelerating 2D Kidney Magnetic Resonance Fingerprinting using Deep Learning Based Tissue Quantification"; J. Magn. Reson. Imaging 2025.
9. X. Zhu, Y. Gao, Z. Xiong, W. Jiang, F. Liu, **H. Sun***; DIP-UP: Deep Image Prior for Unwrapping Phase; Information. 2025; 16 (7):592.

10. M. Li, C. Chen, Z. Xiong, Y. Liu, P. Rong, S. Shan, F. Liu, **H. Sun**, Y. Gao; Quantitative susceptibility mapping via deep neural networks with iterative reverse concatenations and recurrent modules; *Med Phys.* 2025.
11. T. Ding, Y. Gao, Z. Xiong, F. Liu, M.A. Cloos, **H. Sun***; MRF-Mixer: A simulation-based deep learning framework for accelerated and accurate magnetic resonance fingerprinting reconstruction; *Information.* 2023;16(3):218.
12. H. Askari, F. Roosta, **H. Sun***; Training-free medical image inverses via bi-level guided diffusion models; *Proc IEEE/CVF Winter Conf Appl Comput Vis (WACV)*. 2025;75–84.
13. W. Jiang, Z. Xiong, F. Liu, N. Ye, **H. Sun***; Fast Controllable Diffusion Models for Undersampled MRI Reconstruction; 2024 IEEE International Symposium on Biomedical Imaging (ISBI); doi: 10.1109/ISBI56570.2024.10635891.
14. Y. Gao, Z. Xiong, S. Shan, Y. Liu, P. Rong, M. Li, A.H. Wilman, G.B. Pike, F. Liu, **H. Sun***; Plug-and-Play Latent Feature Editing for Orientation-Adaptive Quantitative Susceptibility Mapping Neural Networks; *Medical Imaging Analysis*. 2023.
15. Z. Xiong, Y. Gao, Y. Liu, A. Fazlollahi, P. Nestor, F. Liu, **H. Sun***; Quantitative Susceptibility Mapping through Model-based Deep Image Prior (MoDIP); *NeuroImage*. 2023.
16. W. Zheng, Z. Dai, R. Wu, **H. Sun***; Editorial: Imaging of neurometabolism; *Front Neurosci.* 2023; 17: 1286361.
17. H. Askari, Y. Latif, **H. Sun***; MapFlow: Latent Transition Via Normalizing Flow for Unsupervised Domain Adaptation; *Machine Learning*; 2023.
18. Z. Dai, Z. Yang, Z. Li, M. Li, **H. Sun**, Z. Zhuang, W. Yang, Z. Hu, X. Chen, D. Lin, X. Wu; Increased glymphatic system activity in patients with mild traumatic brain injury; *Front. Neurol. - Applied Neuroimaging*; 2023.
19. S. Shan, Y. Gao, P.Z.Y. Liu, B. Whelan, **H. Sun**, B. Dong, F. Liu, D.E.J. Waddington; Distortion-Corrected Image Reconstruction with Deep Learning on an MRI-Linac; *Magn. Reson. Med.* 2023.
20. Z. Xiong, Y. Gao, F. Liu, **H. Sun***; Affine Transformation Edited and Refined Deep Neural Network for Quantitative Susceptibility Mapping; *NeuroImage*; 2022.
21. N. Nathoo, M. Gee, K. Nelles, J. Burt, **H. Sun**, P. Seres, A.H. Wilman, C. Beaulieu, F. Ba, R. Camicioli; Quantitative susceptibility mapping changes relate to gait issues in Parkinson's Disease; *Can J Neurol Sci.* 2022 Nov 10:1-22. doi: 10.1017/cjn.2022.316.
22. R. Yang, M. Hamilton, **H. Sun**, K. Rawji, S. Sarkar, R. Mirzaei, G.B. Pike, W. Yong, J.F. Dunn; Detecting monocyte trafficking in an animal model of glioblastoma using R2* and quantitative susceptibility mapping; *Cancer Immunology, Immunotherapy*; 2022.
23. D. Nakhid, C. McMorris, **H. Sun**, W.B. Gibbard, C. Tortorelli, C. Lebel; Brain Volume and Magnetic Susceptibility Differences in Children and Adolescents with Prenatal Alcohol Exposure; *Alcoholism: Clinical and Experimental Research*; 2022.
24. A. De, **H. Sun**, D.J. Emery, K.C. Butcher, A.H. Wilman; Quantitative susceptibility-weighted imaging in presence of strong susceptibility sources: Application to hemorrhage; *Magn. Reson Imaging* 2022.
25. Y. Gao, Z. Xiong, A. Fazlollahi, P.J. Nestor, V. Vegh, F. Nasrallah, C. Winter, G.B. Pike, S. Crozier, F. Liu, **H. Sun***; Instant tissue field and magnetic susceptibility mapping from MRI raw phase using Laplacian enhanced deep neural networks; *NeuroImage*. 2022.
26. X. Zhu, Y. Gao, F. Liu, S. Crozier, **H. Sun***; BFRnet: A deep learning-based MR background field removal method for QSM of the brain containing significant pathological susceptibility sources. *Z Med Phys.* 2022.
27. D. Nakhid, C. McMorris, **H. Sun**, B. Gibbard, C. Tortorelli, C. Lebel; Brain iron and mental health symptoms in youth with and without prenatal alcohol exposure; *Nutrients*; 2022.
28. Z. Yang, D. Lin, X. Chen, J. Qiu, S. Li, R. Huang, Z. Yang, **H. Sun**, Y. Liao, J. Xiao, Y.Y. Tang, X. Chen, S. Zhang, Z. Dai; Distinguishing COVID-19 from influenza pneumonia in the early stage through CT imaging and clinical features; *Frontiers in Microbiology*; 2022.
29. Y. Gao, M. Cloos, F. Liu, S. Crozier, G.B. Pike, **H. Sun***; Accelerating quantitative susceptibility and R2* mapping using incoherent undersampling and deep neural network reconstruction. *Neuroimage*. 2021.
30. X. Zhu, Y. Gao, F. Liu, S. Crozier, **H. Sun***; Deep grey matter quantitative susceptibility mapping from small spatial coverages using deep learning. *Z Med Phys.* 2021.
31. Z. Yang, X. Chen, R. Huang, S. Li, D. Lin, Z. Yang, **H. Sun**, G. Liu, J. Qiu, Y. Tang, J. Xiao, Y. Liao, X. Wu, R. Wu, X. Chen, Z. Dai. Atypical presentations of coronavirus disease 2019 (COVID-19) from onset to readmission. *BMC Infect Dis.* 2021 Jan 29;21(1):127.
32. X. Liu, J. Wang, **H. Sun**, S.S.Chandra, S. Croziera, F. Liu; On the regularization of feature fusion and mapping for fast MR multi-contrast imaging via iterative networks; *Magn Reson Imaging*. 2021 Jan 2;77:159-168.
33. Y. Gao, X. Zhu, B.A. Moffat, R. Glarin, A.H. Wilman, G.B. Pike, S. Crozier, F. Liu, **H. Sun***; xQSM: Quantitative Susceptibility Mapping with Octave Convolutional and Noise Regularized Neural Networks; *NMR Biomed.* 2020 Nov 26.
34. Z. Dai, S. Kalra, D. Mah, P. Seres, **H. Sun**, R. Wu, A.H. Wilman; Amide signal intensities may be reduced in the motor cortex and the corticospinal tract of ALS patients; *Eur Radiol.* 2020.
35. M.E. MacDonald, R.J. Williams, D. Rajashekar, R.B. Stafford, A. Hanganu, **H. Sun**, A.J.L. Berman, C.R. McCreary, R. Frayne, N.D. Forkert, G.B. Pike; Age-related differences in cerebral blood flow and cortical thickness with an application to age prediction; *Neurobiol Aging* 2020; 95:131-142.
36. Y. Ma, E.L. Mazerolle, J. Cho, **H. Sun**, Y. Wang, G.B. Pike; Quantification of brain oxygen extraction fraction using QSM and a hyperoxic challenge; *Magn. Reson. Med.* 2020. doi: 10.1002/mrm.28390.

37. X. Chen, Y. Tang, Y. Mo, S. Li, D. Lin, Z. Yang, Z. Yang, **H. Sun**, J. Qiu, Y. Liao, J. Xiao, X. Chen, X. Wu, R. Wu, Z. Dai; A diagnostic model for coronavirus disease 2019 (COVID-19) based on radiological semantic and clinical features: a multi-center study; *Eur Radiol.* 2020 Mar 23.
38. **H. Sun**; Editorial for “Deep-Learning Detection of Cancer Metastasis to the Brain on MRI”; *J. Magn. Reson. Imaging* 2020 Mar 10. doi: 10.1002/jmri.27131.
39. N. Naji, **H. Sun**, A.H. Wilman; On the value of QSM from MP2RAGE for segmenting and quantifying iron-rich deep gray matter; *Magn. Reson. Med.* 2020 Mar 3. doi: 10.1002/mrm.28226.
40. **H. Sun**, J.O. Cleary, R. Glarin, S.C. Kolbe, R.J. Ordidge, B.A. Moffat, G.B. Pike; Extracting more for less: Multi-echo MP2RAGE for simultaneous T1-weighted imaging, T1 mapping, R2* mapping, SWI, and QSM from a single acquisition; *Magn. Reson. Med.* 2020; 83(4):1178-1191.
41. Y. Ma, **H. Sun**, J. Cho, E.L. Mazerolle, Y. Wang, G.B. Pike; Cerebral OEF quantification: A comparison study between quantitative susceptibility mapping and dual-gas calibrated BOLD imaging; *Magn. Reson. Med.* 2020; 83(1):68-82. (selected as **MRM 2020 January Cover Article**)
42. A. De, **H. Sun**, D.J. Emery, K.C. Butcher, A.H. Wilman; Rapid quantitative susceptibility mapping of intracerebral hemorrhage; *J. Magn. Reson. Imaging* 2020; 51(3):712-718.
43. A.M. Elkady, D. Cobzas, **H. Sun**, P. Seres, G. Blevins, A.H. Wilman; Five Year Iron Changes in Relapsing-Remitting Multiple Sclerosis Deep Gray Matter Compared to Healthy Controls; *Mult Scler Relat Disord* 2019; 33:107-115.
44. A.J. Walsh, **H. Sun**, D.J. Emery, A.H. Wilman; Hematocrit Measurement with R2* and Quantitative Susceptibility Mapping in Postmortem Brain; *AJNR Am J Neuroradiol.* 2018; 39(7):1260-1266.
45. **H. Sun**, Y. Ma, M.E. MacDonald, G.B. Pike; Whole Head Quantitative Susceptibility Mapping Using a Least-norm Direct Dipole Inversion Method; *NeuroImage* 2018; 179:166-175.
46. **H. Sun**, A. Klahr, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative Susceptibility Mapping for Following of Intracranial Hemorrhage; *Radiology* 2018; 288(3):830-839.
47. A.M. Elkady, D. Cobzas, **H. Sun**, G. Blevins, A.H. Wilman; Discriminative Analysis of Regional Evolution of Iron and Myelin/Calcium in Deep Gray Matter of Relapsing-Remitting Multiple Sclerosis and Healthy Subjects; *J. Magn. Reson. Imaging* 2018; 48(3):652-668.
48. J. Kmech, E. Fujiwara, D. Cobzas, **H. Sun**, P. Seres, G. Blevins, A.H. Wilman; Cognitive Implications of Deep Gray Matter Iron in Multiple Sclerosis; *AJNR Am J Neuroradiol.* 2017; 38 (5) 942-948.
49. A.M. Elkady, D. Cobzas, **H. Sun**, G. Blevins, A.H. Wilman; Progressive Iron Accumulation Across Multiple Sclerosis Phenotypes Revealed by Sparse Classification of Deep Gray Matter; *J. Magn. Reson. Imaging* 2017; 46(5):1464-1473.
50. M. Juhás, **H. Sun**, M.R.G. Brown, M.B. MacKay, A.H. Wilman, S.M. Dursun, A.J. Greenshaw; Deep Grey Matter Iron Accumulation in Alcohol Use Disorder; *NeuroImage* 2017; 148:115-122.
51. **H. Sun**, P. Seres, A.H. Wilman; Structural and Functional Quantitative Susceptibility Mapping from Standard fMRI Studies; *NMR Biomed.* 2016 Sep 30; doi: 10.1002/nbm.3619.
52. A.M. Elkady, **H. Sun**, A.H. Wilman; Importance of extended spatial coverage for quantitative susceptibility mapping of iron-rich deep grey matter; *J. Magn. Reson. Imaging* 2016; 34(4):574-578.
53. **H. Sun**, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative susceptibility mapping using a superposed dipole inversion method: application to intracranial hemorrhage; *Magn. Reson. Med.* 2016; 76(3):781-91.
54. D. Cobzas, **H. Sun**, A.J. Walsh, R.M. Lebel, G. Blevins, A.H. Wilman; Voxel-based analysis of subcortical grey matter using transverse relaxation and quantitative susceptibility mapping with application to multiple sclerosis; *J. Magn. Reson. Imaging* 2015; 42:1601-1610.
55. **H. Sun**, A.J. Walsh, R.M. Lebel, G. Blevins, I. Catz, J-Q. Lu, E.S. Johnson, D.J. Emery, K.G. Warren, A.H. Wilman; Validation of quantitative susceptibility mapping with Perls’ iron staining for subcortical gray matter; *NeuroImage* 2015; 105:486-492.
56. **H. Sun**, A.H. Wilman; Quantitative susceptibility mapping using single-shot echo-planar imaging; *Magn. Reson. Med.* 2015; 73:1932-1938. (selected as **MRM 2015 May Cover Article**)
57. **H. Sun**, A.H. Wilman; Background field removal using spherical mean value filtering and Tikhonov regularization; *Magn. Reson. Med.* 2014; 71:1151-1157.

NON-PEER-REVIEWED PRE-PRINTS (2 as last author)

1. Z. Wang, W. Dai, T. T. Dao, S. Bollmann, **H. Sun**, C. Engstrom, S. S. Chandra; SAMRI: Segment Anything Model for MRI; *arXiv preprint arXiv:2510.26635*.
2. T. Ding, H. Chen, Y. Gao, Z. Xiong, F. Liu, M.A. Cloos, **H. Sun***; Adaptive Gate-Aware Mamba Networks for Magnetic Resonance Fingerprinting; *arXiv preprint arXiv:2507.03369*.
3. M. Li, C. Chen, Z. Li, Y. Liu, S. Shan, P. Wu, P. Rong, F. Liu, G.B. Pike, A.H. Wilman, **H. Sun**, Y. Gao; SUSEP-Net: Simulation-Supervised and Contrastive Learning-based Deep Neural Networks for Susceptibility Source Separation; *arXiv preprint arXiv:2506.13293*.
4. W. Ma, M. Bran Lorenzana, W. Dai, **H. Sun**, S.S. Chandra; Multi-scale MRI reconstruction via dilated ensemble networks; *p arXiv:2310.04705*.

INVITED TALKS

1. **H. Sun**; AI methods in QSM; ISMRM EMPT 2026 Workshop; Sept. 2026; Annapolis, USA.

2. **H. Sun**; An overview of current deep learning approaches in QSM; ISMRM EMPT Virtual Workshop; Dec. 2025.
3. **H. Sun**; A history of medical imaging; Newcastle Grammar School Medical Engineering Program outreach; Sept. 2024.
4. **H. Sun**; Quantitative MRI using deep learning methods; Invited Visit to Monash Biomedical Imaging Centre, Monash University; July 2024.
5. **H. Sun**; Quantitative MRI and AI applications; Brain Neuromodulation Research Program's Annual Symposium, Hunter Medical Research Institute; June 2024.
6. **H. Sun**; Medical imaging instruments for disease diagnosis, monitoring, and treatment; Guest Lecture for MENG4100: Implants and Assistive Technologies, University of Newcastle; May 2024.
7. **H. Sun**; Diffusion probabilistic models for MRI inverse problems; Australian 7T Network; online; Sept. 2023.
8. **H. Sun**; Decoding magnetic resonance fingerprints with AI; UQ Artificial Intelligence Collaboratory Workshop; Nov. 2022.
9. **H. Sun**; Overview of QSM and deep learning for QSM; ISMRM ANZ QSM Virtual Workshop; Sept. 2022. (**Plenary presentation**)
10. **H. Sun**; Quantitative MRI of the brain with deep learning; EMCR Informal Talks, 1st Meeting; School of ITEE, University of Queensland; July 2022.
11. **H. Sun**; Fast QSM: Accelerated acquisition and instant reconstruction; Centre for Advanced Imaging Annual Symposium 2021; Queensland Brain Institute, Brisbane; Nov. 2021. (**Plenary presentation**)
12. **H. Sun**; Deep learning for MRI acceleration reconstruction; Pattern Recognition and Analysis (COMP3710); University of Queensland, Brisbane; Oct. 2021.
13. **H. Sun**; Advances in New MRI Methods for Human Brain-Quantitative Susceptibility Mapping; University of Queensland, Brisbane; Nov. 2018.
14. **H. Sun**; Whole head quantitative susceptibility mapping using a least-norm direct dipole inversion method; 2018 Alberta Imaging Symposium; Foothills Medical Centre, Calgary; June 2018.
15. **H. Sun**; Quantitative Susceptibility Mapping at high field and ultra-high field MRI; CAI academic visit; Centre for Advanced Imaging, The University of Queensland, Brisbane, Australia; June 2017.
16. **H. Sun**; Quantitative Susceptibility Mapping: reconstruction advances; CSIRO academic visit; Commonwealth Scientific and Industrial Research Organisation, Brisbane, Australia; June 2017.
17. **H. Sun**; Brain iron imaging using MRI Quantitative Susceptibility (QSM); DNDR Seminar Series; The Florey Institute of Neuroscience and Mental Health, The University of Melbourne, Melbourne, Australia; June 2017.
18. **H. Sun**; Quantitative Susceptibility Mapping: Image Reconstruction; CFMM Ultra High Field Winter School, Robarts Research Institute, University of Western Ontario, London; March 2017.
19. **H. Sun**; Brain Iron and Myelin Study Using Quantitative Susceptibility Mapping; NSERC CREATE I3T Advanced Imaging Seminar Series, Foothills Medical Centre, Calgary; Nov. 2016.
20. **H. Sun**; Methods and Applications of Quantitative Susceptibility Mapping; Radiology Research Day of Cumming School of Medicine, Foothills Medical Centre, Calgary; June 2016. (**Plenary presentation**)
21. **H. Sun**; Postmortem validation of QSM for subcortical GM iron quantification in MS; Alberta endMS Retreat 2014: Four Pillars of Multiple Sclerosis Research; Banff, Alberta; Sept. 2014.
22. **H. Sun**; Quantitative Susceptibility Mapping (QSM): Methods and Applications to the Brain; 2014 Alberta Imaging Symposium, University of Alberta, Edmonton; June 2014.
23. **H. Sun**; Susceptibility imaging of the brain at high field; 2013 Alberta Imaging Symposium; Foothills Medical Centre, Calgary; June 2013.

CONTRIBUTED TALKS (8 as first author, 15 as last/corresponding author)

1. W. Jiang, T. Ding, H. Chen, W. Song, Z. Xiong, Y. Gao, N. Ye, F. Liu, **H. Sun***; Self-Supervised Representation Learning of Brain MRI Using DINOv3: From Anatomical Features to Cross-Domain Generalization; ISMRM; Cape Town; May 2026.
2. Z. Chen, Z. Xu, J. Cheng, S. Shan, Z. Yuan, Y. Wang, M. Jiang, G. Zheng, T. Quan, J. Xu, L. Xia, F. Liu, X. Liang, **H. Sun***; BUDA-iQSM+: BUDA Imaging and Deep Learning iQSM+ Enables Rapid and Robust Distortion-free High-Resolution QSM; ISMRM; Cape Town; May 2026.
3. T. Ding, H. Chen, Y. Gao, F. Liu, M. A. Cloos, **H. Sun***; Adaptive Gate-Aware Mamba Networks for Magnetic Resonance Fingerprinting; 2026 IEEE International Symposium on Biomedical Imaging (ISBI), London, UK, April 2026.
4. H. Askari, Y. Luo, **H. Sun***, F. Roosta; Latent Refinement via Flow Matching for Training-free Linear Inverse Problem Solving; NeurIPS; San Diego; December 2025.
5. H. Askari, F. Roosta, **H. Sun***; Training-free medical image inverses via bi-level guided diffusion models; WACV; Tucson, Arizona; March 2025.
6. W. Jiang, Z. Xiong, N. Ye, **H. Sun***; Fast Controllable Diffusion Models for Undersampled MRI Reconstruction; ISBI; Athens, Greece; May 2024.
7. W. Jiang, Y. Gao, **H. Sun***; MR image super-resolution via a variational diffusion model; ISMRM; Toronto; June 2023.
8. W. Jiang, Y. Gao, **H. Sun***; MRI reconstruction using DDPM with sparsely sampled k-space as guidance; ISMRM; Toronto; June 2023.
9. Y. Gao, T. Ding, M.A. Cloos, **H. Sun***; MRF-mixer: a self-supervised deep learning MRF framework; ISMRM; Toronto; June 2023.

10. Z. Xiong, Y. Gao, **H. Sun***; Unsupervised Multi-task learning for solving ill-posed dipole inversion in quantitative susceptibility mapping; QMR Lucca workshop; Lucca, Italy, October 2022.
11. Y. Gao, T. Ding, M.A. Cloos, **H. Sun***; QSM extraction from MRF acquisitions through deep neural networks; QMR Lucca workshop; Lucca, Italy, October 2022.
12. Y. Gao, Z. Xiong, A. Fazlollahi, P.J. Nestor, V. Vegh, G.B. Pike, S. Crozier, F. Liu, **H. Sun***; QSM from the raw phase using an end-to-end neural network; ISMRM; London; May 2022. (**Magna Cum Laude Merit Award**)
13. X. Zhu, Y. Gao, F. Liu, S. Crozier, **H. Sun***; A deep learning-based background field removal method for brains containing high susceptibility sources; ISMRM; London; May 2022.
14. Y. Gao, S. Crozier, F. Liu, **H. Sun***; Accelerating QSM using Compressed Sensing and Deep Neural Network; ISMRM; Vancouver; May 2021.
15. Y. Gao, X. Zhu, S. Crozier, F. Liu, **H. Sun***; xQSM: a deep learning QSM network using Octave Convolution; ISMRM; Sydney; April 2020.
16. **H. Sun**, F. Liu, S. Crozier; Differentiating hemorrhage, calcification, and edema of glioblastoma using quantitative susceptibility mapping (QSM); AUS MR in RT workshop; Noosa; 2019.
17. **H. Sun**, M.E. MacDonald, E.L. Mazerolle, K. Sabourin, G.B. Pike; Localization of GPi for MRgFUS pallidotomy: a comparison between high-resolution FGATIR, R2* and QSM at 3 T; ISMRM; Montreal; May 2019. (**Magna Cum Laude Merit Award**)
18. F. Schweser, W. Li, **H. Sun**, D. Zhou, N. Bertolino, P. Polak, Y. Wang, A.H. Wilman, K. Bredies, R. Zivadinov, S.D. Robinson; An illustrated comparison of background field elimination methods for phase MRI and QSM; ISMRM; Singapore; May 2016.
19. **H. Sun**, P. Seres, A.H. Wilman; Subcortical Grey Matter Susceptibility Mapping from Standard fMRI studies; ISMRM; Toronto; June 2015.
20. A.M. Elkady, **H. Sun**, A.J. Walsh, G. Blevins, Z. Dai, A.H. Wilman; Quantitative Susceptibility Mapping of Lesions in Multiple Sclerosis; ISMRM; Toronto; June 2015.
21. **H. Sun**, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; QSM of intracranial hemorrhage: Artifact reduction using a superposition method; QSM workshop; Duke University, Durham; Oct. 2014.
22. **H. Sun**, V.V. Divekar, M. Kate, L.C. Gioia, C.A. Baron, C. Beaulieu, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative Susceptibility Mapping of Intracranial Hemorrhage: Clinical Results and Numerical Simulation; ISMRM; Milan; May 2014. (**Magna Cum Laude Merit Award**)
23. **H. Sun**, A.H. Wilman; Echo-Planar Susceptibility Mapping: QSM using single-shot gradient EPI; ISMRM; Milan; May 2014. (**Summa Cum Laude Merit Award**)
24. **H. Sun**, A.J. Walsh, G. Blevins, J-Q. Lu, E.S. Johnson, K.G. Warren, A.H. Wilman; Susceptibility mapping development and application to brain iron in multiple sclerosis using postmortem studies; QSM workshop; Cornell University, Ithaca; July 2013.
25. **H. Sun**, A.H. Wilman; Susceptibility mapping using Tikhonov regularization enabled harmonic artifact removal; ISMRM; Salt Lake City; April 2013. (**Summa Cum Laude Merit Award**)

PUBLISHED ABSTRACTS (33 as first author, 37 as last/corresponding author, 71 in ISMRM)

1. T. Ding, H. Chen, Y. Gao, P. Wu, Z. Xiong, S. Shan, F. Liu, M. A. Cloos, **H. Sun***; Mamba-MRF: A Deep Mamba Network for Highly Accelerated Magnetic Resonance Fingerprinting Reconstruction; ISMRM; Cape Town; May 2026.
2. Z. Chen, Z. Xu, J. Cheng, S. Shan, Z. Yuan, Y. Wang, M. Jiang, G. Zheng, T. Quan, J. Xu, L. Xia, F. Liu, X. Liang, **H. Sun***; BUDA-iQSM+: BUDA Imaging and Deep Learning iQSM+ Enables Rapid and Robust Distortion-free High-Resolution QSM; ISMRM; Cape Town; May 2026.
3. A. Gao, Y. Liu, Y. Liu, S. Shan, R. Zhou, P. Wu, F. Liu, G. Bruce Pike, **H. Sun**, Y. Gao; PDDF-Net: A deep neural network for diagnosing Parkinson's Disease using QSM and T1w images; ISMRM; Cape Town; May 2026.
4. S. Prima, Y. Gao, **H. Sun***; A Self-Supervised Transformer for Myelin Water Fraction Mapping from Multi-Echo Gradient-Echo Imaging; ISMRM; Cape Town; May 2026.
5. W. Jiang, T. Ding, H. Chen, W. Song, Z. Xiong, Y. Gao, N. Ye, F. Liu, **H. Sun***; Self-Supervised Representation Learning of Brain MRI Using DINOv3: From Anatomical Features to Cross-Domain Generalization; ISMRM; Cape Town; May 2026.
6. J. F. Dunn, T. Makarowski, A. Palset, Y. Wu, **H. Sun**, Y. Yee, G. A. Devenyi; Investigating magnetic susceptibility changes caused by hypoxia and myelin in the mouse brain at 9.4T using QSM and R2*; ISMRM; Cape Town; May 2026.
7. S. Prima, Z. Xiong, A.H. Wilman, **H. Sun***; DeepRelaxo: A Generalizable Self-supervised Method for Brain R2 Mapping; ISMRM; Honolulu; May 2025.
8. T. Ding, Y. Gao, Z. Xiong, F. Liu, M. Cloos, **H. Sun***; Partial Volume Estimation from MRF Acquisition Using a Deep Learning Approach; ISMRM; Honolulu; May 2025.
9. W. Jiang, W. Song, Y. Gao, N. Ye, F. Liu, **H. Sun***; Consistent MRI Reconstruction with Diffusion Models and Sequential Monte Carlo; ISMRM; Honolulu; May 2025.

10. Z. Xiong, Y. Gao, W. Jiang, K. Butcher, A. Wilman, **H. Sun***; Rapid Quantitative Susceptibility Mapping for Intracranial Hemorrhage Using Deep Learning-based 2.5D Diffusion Models; ISMRM; Honolulu; May 2025.
11. **H. Sun**, T. Islam, M. Barth, Z. Chen; Synthesising 3T DWI from ultra-low-field (64mT) acquisitions using generative diffusion models; ISMRM; Singapore; May 2024.
12. Z. Xiong, Y. Gao, **H. Sun***; Quantitative Susceptibility Mapping by Controllable Diffusion Models; ISMRM; Singapore; May 2024.
13. W. Jiang, F. Liu, N. Ye, **H. Sun***; Robustness of Diffusion Model-Based Methods to Distribution Shifts in Medical Imaging; ISMRM; Singapore; May 2024.
14. X. Zhu, Y. Gao, Z. Xiong, W. Jiang, F. Liu, S. Crozier, **H. Sun***; A deep image prior based refinement for 3D phase unwrapping in brain MRI; ISMRM; Singapore; May 2024.
15. T. Ding, Y. Gao, Z. Xiong, M. Cloos, **H. Sun***; Multi Complex-valued Spatio-temporal Fusion Networks for Robust MRF Reconstruction; ISMRM; Singapore; May 2024.
16. Y. Gao, Z. Xiong, S. Shan, M. Li, A.H. Wilman, G.B. Pike, F. Liu, **H. Sun***; QSM Reconstruction of Arbitrary Dipole Orientations using an End-to-end Neural Network via Latent Feature Editing; ISMRM; Singapore; May 2024.
17. C. Chen, Y. Gao, M. Li, Z. Xiong, F. Liu, **H. Sun***; LoopNet: A New Baseline Network for QSM Dipole Inversion; ISMRM; Singapore; May 2024.
18. T. Makarowski, **H. Sun**, J.F. Dunn; Comparing R2* and QSM at 9.4T for the ability to detect increased deoxyhemoglobin (hypoxia) in the mouse brain; ISMRM; Singapore; May 2024.
19. W. Jiang, F. Liu, N. Ye, **H. Sun***; A Study on Diffusion Models for Domain Shifts in MRI Reconstruction; ISMRM ANZ Chapter; Brisbane; Nov 2023.
20. Z. Xiong, Y. Gao, **H. Sun***; Quantitative Susceptibility Mapping by Controllable Diffusion Models; ISMRM ANZ Chapter; Brisbane; Nov 2023.
21. T. Ding, Y. Gao, Z. Xiong, M. Cloos, **H. Sun***; Complex-valued Spatio-temporal Fusion Network for Robust MRF Reconstruction; ISMRM ANZ Chapter; Brisbane; Nov 2023.
22. S. Prima, Z. Xiong, **H. Sun***; Deep Learning-based Brain R2* Mapping from Variable Acquisition Parameters; ISMRM ANZ Chapter; Brisbane; Nov 2023.
23. S. Prima, Z. Xiong, **H. Sun***; Brain MRI R2* Quantification using Deep Neural Networks; EMBC; Sydney; July 2023.
24. T. Ding, Y. Gao, **H. Sun***; A complex-valued deep neural network for Magnetic Resonance Fingerprinting; EMBC; Sydney; July 2023.
25. W. Jiang, Y. Gao, **H. Sun***; MR image super-resolution via a variational diffusion model; ISMRM; Toronto; June 2023.
26. W. Jiang, Y. Gao, **H. Sun***; MRI reconstruction using DDPM with sparsely sampled k-space as guidance; ISMRM; Toronto; June 2023.
27. Y. Gao, T. Ding, M.A. Cloos, **H. Sun***; MRF-mixer: a self-supervised deep learning MRF framework; ISMRM; Toronto; June 2023.
28. Z. Xiong, Y. Gao, **H. Sun***; Real-time style transfer for quantitative susceptibility mapping using unsupervised learning; ISMRM; Toronto; June 2023.
29. Q. Shafqat, Y. Wu, U.J. Ohazukosi, T. Makarowski, **H. Sun**, R. Tariq, J.F. Dunn; Systemic inflammation causes cerebral hypoperfusion, reductions in brain susceptibility and hippocampal hypoxia: a 9.4T MRI animal study; ISMRM; Toronto; June 2023.
30. R. Tariq, R. Muhammed, Y. Wu, Q. Shafqat, **H. Sun**, J.F. Dunn; Magnetic Susceptibility vs Relaxometry in the Characterization of an Animal Model of Multiple Sclerosis; ISMRM; Toronto; June 2023.
31. S. Shan, Y. Gao, P. Liu, T. Reynolds, B. Dong, **H. Sun**, M Li, G Liney, F Liu, P Keall, D Waddington; Motion-corrected Image Reconstruction Using an Unrolling Network on an MRI-Linac; AUS MR in RT workshop; Noosa; 2022.
32. R. Steele, Q. Shafqat, Y. Wu, **H. Sun**, J.F. Dunn; Cerebral blood flow and magnetic susceptibility are reduced in the bacterial lipopolysaccharide (LPS) mouse model of inflammation; Alberta Imaging Symposium; Foothills Medical Centre, Calgary; 2022.
33. Z. Xiong, Y. Gao, **H. Sun***; Unsupervised Multi-task learning for solving ill-posed dipole inversion in quantitative susceptibility mapping; QMR Lucca workshop; Lucca, Italy, October 2022.
34. Z. Xiong, Y. Gao, **H. Sun***; Swin-QSM: Quantitative susceptibility mapping using Swin-Transformer; QMR Lucca workshop; Lucca, Italy, October 2022.
35. Y. Gao, T. Ding, M.A. Cloos, **H. Sun***; QSM extraction from MRF acquisitions through deep neural networks; QMR Lucca workshop; Lucca, Italy, October 2022.
36. D. Nakhid, C. McMorris, **H. Sun**, W.B. Gibbard, C. Tortorelli, C. Lebel; Magnetic susceptibility and anxiety symptoms in children with and without prenatal alcohol exposure; OHBM; Glasgow; 2022.
37. Y. Gao, Z. Xiong, A. Fazlollahi, P.J. Nestor, V. Vegh, G.B. Pike, S. Crozier, F. Liu, **H. Sun***; QSM from the raw phase using an end-to-end neural network; ISMRM; London; May 2022.
38. X. Zhu, Y. Gao, F. Liu, S. Crozier, **H. Sun***; A deep learning-based background field removal method for brains containing high susceptibility sources; ISMRM; London; May 2022.
39. Z. Xiong, Y. Gao, S. Bollmann, **H. Sun***; A deep learning dipole inversion method for QSM of arbitrary head orientation and image resolution; ISMRM; London; May 2022.

40. Q. Shafqat, R. Muhammed, **H. Sun**, Y. Wu, A.M. Hamilton, M. Hashem, J.F. Dunn; Decrease in magnetic susceptibility correlates with reduction in cortical blood flow in a model of systemic inflammation: A 9.4T in-vivo study; ISMRM; London; May 2022.
41. S. Shan, Y. Gao, P. Liu, B. Whelan, D. Waddington, **H. Sun**, F. Liu, P. Keall; DFReconNet: Distortion Free Image Reconstruction using Deep Neural Network for MRI-Linac; ISMRM; London; May 2022.
42. S. Shan, Y. Gao, P. Liu, T. Reynolds, B. Dong, **H. Sun**, M. Li, G. Liney, F. Liu, P. Keall, D. Waddington; Motion-corrected Image Reconstruction with Unrolling Networks on an MRI-Linac; American Association of Physicists in Medicine (AAPM); Washington; 2022.
43. Z. Xiong, Y. Gao, **H. Sun***; Deep learning QSM for acquisitions of arbitrary orientation; ISMRM ANZ Chapter; online; Nov 2021.
44. Y. Gao, F. Liu, **H. Sun***; Universal QSM Reconstruction from Raw Phase using Deep Neural Networks; ISMRM ANZ Chapter; online; Nov 2021.
45. Y. Gao, S. Crozier, F. Liu, **H. Sun***; Accelerating QSM using Compressed Sensing and Deep Neural Network; ISMRM; Vancouver; 2021.
46. D. Nakhid, **H. Sun**, C. McMorris, C. Lebel; Brain Volume and Susceptibility Differences in Children with Prenatal Alcohol Exposure; International Society for Developmental Psychobiology; Rockville; 2020.
47. **H. Sun**, M.E. MacDonald, R.M. Lebel, G.B. Pike; Simultaneous T1-weighted imaging, R2* mapping, and QSM from a multi-echo MP2RAGE sequence using a radial fan-beam sampling scheme at 3 Tesla; ISMRM; Sydney; 2020.
48. Y. Gao, X. Zhu, S. Crozier, F. Liu, **H. Sun***; xQSM: a deep learning QSM network using Octave Convolution; ISMRM; Sydney; 2020.
49. X. Zhu, Y. Gao, F. Liu, S. Crozier, **H. Sun***; Robust deep grey matter QSM from small brain coverages using a deep learning-based method; ISMRM; Sydney; 2020.
50. X. Liu, J. Wang, F. Tang, **H. Sun**, F. Liu, S. Crozier; Rapid Region-of-Interest MRI Reconstruction Using Context-Aware Non-Local U-Net; ISMRM; Sydney; 2020.
51. A. De, **H. Sun**, K.C. Butcher, A.H. Wilman; Application of Fourier-domain Analysis Based Unwrapping Technique in Quantitative Susceptibility Mapping (QSM) of Intracerebral Hemorrhage; ISMRM; Sydney; 2020.
52. S. Scott, M.E. MacDonald, D. Rajashekar, W-Q Liu, **H. Sun**, G.B. Pike; Clustering Analysis of Multiple Sclerosis Lesions with T1-&T2-weighted, Diffusion, Quantitative Susceptibility Mapping, and Magnetization Transfer Ratio Imaging; OHBM; Montreal; 2020.
53. Y. Ma, E.L. Mazerolle, J. Cho, **H. Sun**, Y. Wang, G.B. Pike; Quantification of Cerebral Oxygen Extraction Fraction (OEF) Using QSM with a Hyperoxic Challenge; OHBM; Montreal; 2020.
54. **H. Sun**, F. Liu, S. Crozier; Differentiating hemorrhage, calcification, and edema of glioblastoma using quantitative susceptibility mapping (QSM); AUS MR in RT workshop; Noosa; 2019.
55. J.O. Cleary, **H. Sun**, R. Glarin, S.C. Kolbe, R.J. Ordidge, B.A. Moffat, G.B. Pike; Extracting more for less at Ultra High Field: Multi-echo MP2RAGE for simultaneous T1-weighted imaging, T1 mapping, R2* mapping, SWI, and QSM in a single acquisition; British Society of Neuroradiologists Annual Meeting; Cardiff; 2019.
56. Y. Gao, X. Zhu, S. Bollmann, M. Barth, S. Crozier, F. Liu, **H. Sun***; OctQSM - A deep learning QSM method with Octave convolution; QSM workshop; Seoul; 2019.
57. X. Zhu, Y. Gao, S. Bollmann, M. Barth, F. Liu, S. Crozier, **H. Sun***; Comparison of Conventional and Learning Based QSM on Spatial Coverage Effect; QSM workshop; Seoul; 2019.
58. **H. Sun**, M.E. MacDonald, E.L. Mazerolle, K. Sabourin, G.B. Pike; Localization of GPi for MRgFUS pallidotomy: a comparison between high-resolution FGATIR, R2* and QSM at 3 T; ISMRM; Montreal; 2019.
59. Y. Ma, **H. Sun**, J. Cho, E.L. Mazerolle, Y. Wang, G.B. Pike; Whole-brain OEF quantification: a comparison study between QSM and dual-gas calibrated BOLD; ISMRM; Montreal; 2019.
60. Elkady, D. Cobzas, **H. Sun**, G. Blevins, P. Seres, A.H. Wilman; Five Year Changes in Iron and Myelin in Relapsing-Remitting Multiple Sclerosis Deep Gray Matter Compared to Healthy Controls; ISMRM; Montreal; 2019.
61. Z. Dai, S. Kalra, D. Mah, P. Seres, G. Yan, **H. Sun**, Z. Shen, R. Wu, A.H. Wilman; Amide proton transfer in amyotrophic lateral sclerosis; ISMRM; Montreal; 2019.
62. Z. Dai, H. Zhang, A.H. Wilman, H. Xu, G. Xiao, **H. Sun**, Zhuang, Y. Jia, Z. Shen, G. Yan, R. Wu; Glutamate imaging in schizophrenia prodrome; ISMRM; Montreal; 2019.
63. M.E. MacDonald, W. Liu, S. Scott, C. Rockel, D. Rajashekar, J.L. Specht, **H. Sun**, G.B. Pike; White Matter Tract-Defined Lesion Loads in Relapsing-Remitting Multiple Sclerosis; ISMRM; Montreal; 2019.
64. **H. Sun**, G.B. Pike; Whole head quantitative susceptibility mapping using a least-norm direct dipole inversion method; 2018 Alberta Imaging Symposium; Foothills Medical Centre, Calgary; 2018.
65. S.E. Ha, R.G. Sah, **H. Sun**, P.A. Barber; Quantifying brain iron deposition in patients with Transient Ischemic Attack and Healthy Control using quantitative susceptibility mapping; HBI Summer Student Symposium; Calgary; 2018.
66. **H. Sun**, Y. Ma, M.E. MacDonald, J.O. Cleary, S. Kolbe, G.B. Pike; QSM of the Entire Head Using a Least-norm Direct Dipole Inversion Method; ISMRM Neuro Workshop; Seoul; 2018.
67. **H. Sun**, J.O. Cleary, R. Glarin, Y. Ma, K. O'Brien, S.C. Kolbe, B.A. Moffat, R.J. Ordidge, G.B. Pike; Multi-echo MP2RAGE at 7T enables simultaneous T1w, quantitative T1, T2* and QSM from a single acquisition; ISMRM Neuro Workshop; Seoul; 2018.

68. **H. Sun**, J.O. Cleary, R. Glarin, S.C. Kolbe, B.A. Moffat, R.J. Ordidge, G.B. Pike; Multi-channel phase combination for quantitative susceptibility mapping at 7T using Phase-Offsets Estimation from Multi-echoes (POEM) method; ISMRM; Paris; 2018.
69. **H. Sun**, J. O. Cleary, R. Glarin, Y. Ma, K. O'Brien, S.C. Kolbe, B.A. Moffat, R.J. Ordidge, G.B. Pike; Extracting more: Multi-echo MP2RAGE at 7T enables simultaneous T1w, quantitative T1, T2* and susceptibility mapping from a single acquisition; ISMRM; Paris; 2018.
70. De, **H. Sun**, A. Elkady, D.J. Emery, K. Butcher, A.H. Wilman; Rapid Quantitative Susceptibility Mapping of Intracranial Hemorrhage using Echo Planar Imaging; ISMRM; Paris; 2018.
71. De, **H. Sun**, A. Elkady, P. Seres, A.H. Wilman; Effects of Motion in Quantitative Susceptibility Mapping of Brain; ISMRM; Paris; 2018.
72. J.O. Cleary, **H. Sun**, R. Glarin, P. Yoo, B.A. Moffat, R.J. Ordidge, S.C. Kolbe; COSMOS for Estimating Variation in Single Orientation Quantitative Susceptibility Mapping of the Brain: An Ultra High Field Study; ISMRM; Paris; 2018.
73. Elkady, D. Cobzas, **H. Sun**, G. Blevins, A.H. Wilman; Discriminative Analysis of Regional Evolution of Iron and Myelin/Calcium in Deep Gray Matter of Multiple Sclerosis and Healthy Subjects; ISMRM; Paris; 2018.
74. R. Yang, A.M. Hamilton, **H. Sun**, R. Mirzaei, S. Sarkar, G.B. Pike, V.W. Yong, J.F. Dunn; Comparison of R2* and quantitative susceptibility mapping in the characterizing tumor hypoxia in a mouse model of glioblastoma; ISMRM; Paris; 2018.
75. P.B. Najafabadi, A. Klahr, **H. Sun**, A. Elkady, D.J. Emery, K. Butcher, A.H. Wilman; QSM in stroke: Veins, Tissue and Cerebral Microbleeds; ISMRM; Paris; 2018.
76. M.E. MacDonald, N.D. Forkert, A. Hanganu, Y. Ma, R.J. Williams, **H. Sun**, R. Stafford, C. McCreary, R. Frayne, G.B. Pike; Cerebrovascular Brain Aging Examined with Arterial Spin Labelling and Applied to Age Prediction; ISMRM; Paris; 2018.
77. **H. Sun**, Y. Ma, M.E. MacDonald, G.B. Pike; Tikhonov regularization aided quantitative susceptibility mapping of whole brain without background field removal; ISMRM; Honolulu; 2017.
78. **H. Sun**, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative Susceptibility Mapping of Intracranial Hemorrhage: Time Evolution and Comparison to CT; ISMRM; Honolulu; 2017.
79. A.M. Elkady, D. Cobzas, **H. Sun**, G. Blevins, A.H. Wilman; Progressive Iron Accumulation in Multiple Sclerosis Phenotypes Revealed by Sparse Classification of Deep Gray Matter; ISMRM; Honolulu; 2017.
80. F. Schweser, W. Li, **H. Sun**, D. Zhou, N. Bertolino, P. Polak, Y. Wang, A.H. Wilman, K. Bredies, R. Zivadinov, S.D. Robinson; Quantitative comparison of background field elimination methods in a realistic numerical model; QSM workshop; Medical University of Graz, Austria; 2016.
81. **H. Sun**, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; Time evolution of QSM in Intracranial Hemorrhage; QSM workshop; Medical University of Graz, Austria; 2016.
82. **H. Sun**, M.E. MacDonald, Y. Ma, G.B. Pike; Regularization-aided susceptibility inversion without background field removal; QSM workshop; Medical University of Graz, Austria; 2016.
83. M. Juhás, **H. Sun**, M.R.G. Brown, M.B. MacKay, J. Benoit, E. Dametto, A.H. Wilman, A.J. Greenshaw, S.M. Dursun; Brain Iron in Alcohol Use Disorder; Alberta Imaging Symposium; Foothills Medical Centre, Calgary; 2016.
84. S. Treit, **H. Sun**, P. Seres, A.H. Wilman, C. Beaulieu; Quantitative Susceptibility Mapping of White Matter Identified by Diffusion Tensor Tractography; Organization for Human Brain Mapping; Geneva; 2016.
85. F. Schweser, W. Li, **H. Sun**, D. Zhou, N. Bertolino, P. Polak, Y. Wang, A.H. Wilman, K. Bredies, R. Zivadinov, S.D. Robinson; An illustrated comparison of background field elimination methods for phase MRI and QSM; ISMRM; Singapore; 2016.
86. **H. Sun**, M.E. MacDonald, G.B. Pike; Phase Correction of a Bipolar Gradient-Echo Acquisition for Quantitative Susceptibility Mapping; ISMRM; Singapore; 2016.
87. **H. Sun**, A.H. Wilman; Clinical Applications of Susceptibility Mapping in the Brain; 6th Annual Graduate Research Symposium; University of Alberta, Edmonton; 2015.
88. **H. Sun**, P. Seres, A.H. Wilman; Subcortical Grey Matter Susceptibility Mapping from Standard fMRI studies; ISMRM; Toronto; 2015.
89. **H. Sun**, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative Susceptibility Mapping of Intracranial Hemorrhage: Artifacts Reduction; ISMRM; Toronto; 2015.
90. A.M. Elkady, **H. Sun**, A.H. Wilman; Limitations of Accelerated QSM by FOV Restriction to Deep Gray Matter; ISMRM; Toronto; 2015.
91. A.M. Elkady, **H. Sun**, A.J. Walsh, G. Blevins, Z. Dai, A.H. Wilman; Quantitative Susceptibility Mapping of Lesions in Multiple Sclerosis; ISMRM; Toronto; 2015.
92. D. Cobzas, **H. Sun**, A.J. Walsh, R.M. Lebel, G. Blevins, A.H. Wilman; Voxel-based analysis of subcortical grey matter using transverse relaxation and quantitative susceptibility mapping: application to multiple sclerosis; ISMRM; Toronto; 2015.
93. Z. Dai, Y. Jia, G. Yan, F. Duan, G. Xiao, Z. Shen, **H. Sun**, A.H. Wilman, R. Wu; pH-weighted imaging in diabetes mellitus suffering acute cerebral ischemic stroke; ISMRM; Toronto; 2015.
94. **H. Sun**, R. Topfer, A.H. Wilman; Quantitative susceptibility mapping using MRI; Canadian Graduate Engineering Consortium Recruiting Event; University of Alberta, Edmonton; 2014.

95. **H. Sun**, M. Kate, L.C. Gioia, D.J. Emery, K. Butcher, A.H. Wilman; QSM of intracranial hemorrhage: artifact reduction using a superposition method; QSM workshop; Duke University, Durham; 2014.
96. **H. Sun**, A.J. Walsh, R.M. Lebel, G. Blevins, I. Catz, J-Q. Lu, E.S. Johnson, D.J. Emery, K.G. Warren, A.H. Wilman; Postmortem validation of quantitative susceptibility mapping for subcortical gray matter iron quantification in multiple sclerosis; Alberta endMS Retreat; Banff, Alberta; 2014.
97. **H. Sun**, A.H. Wilman; Echo-Planar Susceptibility Mapping; ISMRM; Milan; 2014.
98. **H. Sun**, V.V. Divekar, M. Kate, L.C. Gioia, C.A. Baron, C. Beaulieu, D.J. Emery, K. Butcher, A.H. Wilman; Quantitative Susceptibility Mapping of Intracranial Hemorrhage: Clinical Results and Numerical Simulation; ISMRM; Milan; 2014.
99. **H. Sun**, A.H. Wilman; Quantitative susceptibility mapping (QSM): methods and applications to the brain; 2014 Alberta Imaging Symposium, University of Alberta, Edmonton; 2014.
100. **H. Sun**, A.H. Wilman; Brain iron mapping in a few seconds using MRI; 5th Annual Graduate Research Symposium; University of Alberta, Edmonton; 2014.
101. **H. Sun**, A.J. Walsh, G. Blevins, J-Q. Lu, E.S. Johnson, K.G. Warren, A.H. Wilman; Susceptibility mapping development and application to brain iron in multiple sclerosis using postmortem studies; QSM workshop; Cornell University, Ithaca; 2013.
102. **H. Sun**, A.H. Wilman; Susceptibility imaging of the brain at high field; 2013 Alberta Imaging Symposium; Foothills Medical Centre, Calgary; 2013.
103. **H. Sun**, A.J. Walsh, K.G. Warren, G. Blevins, E.S. Johnson, J-Q. Lu, A.H. Wilman; Validation of QSM for brain iron mapping in multiple sclerosis using postmortem studies; QSM workshop; Cornell University, Ithaca; 2013.
104. R. Topfer, **H. Sun**, A.H. Wilman; Edge-Extended Harmonic Phase Processing Incorporating Priors; QSM workshop; Cornell University, Ithaca; 2013.
105. **H. Sun**, A.J. Walsh, R.M. Lebel, G. Blevins, I. Catz, J-Q. Lu, E.S. Johnson, D.J. Emery, K.G. Warren, A.H. Wilman; Validation of susceptibility mapping for quantification of iron in subcortical grey matter in multiple sclerosis; ISMRM; Salt Lake City; 2013.
106. **H. Sun**, A.H. Wilman; Susceptibility Mapping Using Regularization Enabled Harmonic Artifact Removal; ISMRM; Salt Lake City; 2013.
107. **H. Sun**, A.J. Walsh, K.G. Warren, G. Blevins, E.S. Johnson, J-Q. Lu, A.H. Wilman; Quantitative susceptibility mapping for multiple sclerosis using MRI; 4th Annual Graduate Research Symposium; University of Alberta, Edmonton; 2013.
108. **H. Sun**, A.H. Wilman; Development of Quantitative susceptibility mapping; Canadian Graduate Engineering Consortium Recruiting Event; University of Alberta, Edmonton; 2013.
109. **H. Sun**, A.H. Wilman; Quantitative Susceptibility Mapping at 4.7T MRI; 3rd Annual Graduate Research Symposium; University of Alberta, Edmonton; 2012.
110. **H. Sun**, A.H. Wilman; Quantitative Susceptibility Mapping from Multi-echo Gradient-echo Sequence at 4.7 T; Alberta Imaging Symposium; Foothills Medical Centre, Calgary; 2012.

DOCTORAL THESIS

- **H. Sun**; Quantitative Susceptibility Mapping in Human Brain: Methods Development and Applications; Ph.D. thesis; Department of Biomedical Engineering, University of Alberta; September 2015.